Bent Marshak Waves: Axisymmetric Cylindrical Model

Eigenvalues, Longitudinal Calibration, and Front Curvature

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1 Physical setup and assumptions

We consider an optically thick, stationary foam within a cylindrical sleeve of radius R. The axial direction is z, the radial direction is $r \in [0, R]$, and axial symmetry implies $\partial_\theta = 0$. A radiation-temperature drive $T_s(t)$ is imposed at $z = 0$. We take constant foam density ρ and constant Rosseland opacity κ; the wall has albedo a. Let

$$U(z, r, t) = T^4(z, r, t), \qquad U_s(t) = T_s^4(t). \tag{1.1}$$

As in the quasi-steady region of Hurricane and Hammer's *Bent Marshak Waves* (2006), the temperature field is treated as harmonic behind a steep front. This is not a full time-dependent diffusion solution, and corrections to the front position should be distinguished from a self-consistent moving-boundary solution.

2 Governing equation and cylindrical modes

The diffusion approximation gives

$$\rho \frac{\partial e}{\partial t} = \frac{4}{3} \nabla \cdot \left[\frac{\sigma}{\rho \kappa} \nabla U \right], \tag{2.1}$$

where e is *specific* internal energy (energy per mass) and σ is the Stefan–Boltzmann constant in the paper's units. Assuming $\partial_t e$ is negligible away from the front, and constant ρ, κ,

$$\boxed{\partial_z^2 U + \frac{1}{r} \partial_r (r \partial_r U) = 0.} \tag{2.2}$$

With $U = Z(z)F(r)$, separation yields

$$Z'' - k^2 Z = 0, \qquad F'' + \frac{1}{r} F' + k^2 F = 0. \tag{2.3}$$

The finite radial solution is $F(r) = J_0(kr)$; $Y_0(kr)$ is singular on the axis. The full modal solution is

$$\boxed{U(z, r, t) = \sum_{n=0}^{\infty} J_0(k_n r) \left[A_n(t) e^{k_n z} + B_n(t) e^{-k_n z} \right].} \tag{2.4}$$

3 Wall boundary condition and eigenvalues

The axis is regular, $\partial_r U(0,z,t)=0$. The outward radiative loss at $r=R$ gives the Robin condition

$$\partial_r U|_R = -hU(R,z,t), \qquad h=\frac{3}{4}\rho\kappa(1-a). \tag{3.1}$$

Introduce $\varepsilon = hR = \frac{3}{4}\rho\kappa R(1-a)$ and $\lambda_n = k_n R$. Since $J_0'(x) = -J_1(x)$, each mode obeys

$$\boxed{\lambda_n J_1(\lambda_n) = \varepsilon J_0(\lambda_n), \qquad k_n = \lambda_n/R.} \tag{3.2}$$

This replaces the planar relation $\tan(k_n L) = \varepsilon/(k_n L)$. For $\varepsilon > 0$, the roots $\lambda_n > 0$ are ordered increasingly; $\lambda_0 \to 0$ as $\varepsilon \to 0$.

3.1 Small-wall-loss eigenvalue expansions

The Bessel series are

$$J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} + O(x^6), \tag{3.3}$$

$$J_1(x) = \frac{x}{2} - \frac{x^3}{16} + \frac{x^5}{384} + O(x^7). \tag{3.4}$$

For the fundamental mode,

$$\varepsilon = \frac{\lambda_0 J_1(\lambda_0)}{J_0(\lambda_0)} = \frac{\lambda_0^2}{2} + \frac{\lambda_0^4}{16} + \frac{\lambda_0^6}{96} + O(\lambda_0^8), \tag{3.5}$$

so

$$\boxed{\lambda_0 = \sqrt{2\varepsilon}\left(1 - \frac{\varepsilon}{8} + \frac{5\varepsilon^2}{384} + O(\varepsilon^3)\right).} \tag{3.6}$$

For $n \geq 1$, let $\alpha_n = j_{1,n}$ be the n-th positive zero of J_1 and write $\lambda_n = \alpha_n + \delta_n$. Expanding (3.2) around α_n and using $J_1'(\alpha_n) = J_0(\alpha_n)$ gives

$$\boxed{\lambda_n = \alpha_n + \frac{\varepsilon}{\alpha_n} - \frac{\varepsilon^2}{2\alpha_n^3} + O(\varepsilon^3), \quad n \geq 1.} \tag{3.7}$$

The first positive zeros are $\alpha_1 \simeq 3.831706$, $\alpha_2 \simeq 7.015587$, $\alpha_3 \simeq 10.173468$.

3.2 Weighted orthogonality and exact source coefficients

Define $\phi_n(r) = J_0(k_n r)$. The eigenfunctions are orthogonal with cylindrical weight r:

$$\int_0^R r\phi_n(r)\phi_m(r)\,dr = N_n\delta_{nm}, \qquad N_n = \frac{R^2}{2}\left[J_0^2(\lambda_n) + J_1^2(\lambda_n)\right]. \tag{3.8}$$

The simpler norm using only J_1 applies to Dirichlet zeros of J_0 and must *not* be used for the present Robin eigenvalues. For a radially uniform boundary $U(0,r,t) = U_s(t)$, let $1 = \sum_{n\geq 0} b_n\phi_n(r)$. Then

$$\boxed{b_n = \frac{2J_1(\lambda_n)}{\lambda_n[J_0^2(\lambda_n) + J_1^2(\lambda_n)]}.} \tag{3.9}$$

For $\varepsilon \ll 1$, $b_0 = 1 + \varepsilon/4 + O(\varepsilon^2)$, while $b_{n\geq 1} = O(\varepsilon)$. The source cannot be represented *exactly* by the fundamental mode alone for nonzero wall loss.

4 Longitudinal conditions and quasi-steady temperature profile

The source condition is $U(0,r,t) = U_s(t)$. First solve an auxiliary *flat-front* boundary problem with a reference plane $z = \ell(t)$ and $U(\ell,r,t) = 0$. Modal separation and (3.9) then give

$$\boxed{\frac{U(z,r,t)}{U_s(t)} = \sum_{n=0}^{\infty} b_n J_0(k_n r) \frac{\sinh[k_n(\ell(t)-z)]}{\sinh[k_n \ell(t)]}.} \tag{4.1}$$

This field meets both longitudinal boundary conditions (up to the usual convergence interpretation at the source–wall corner). The leading-mode approximation is

$$\frac{U}{U_s} \simeq b_0 J_0(k_0 r) \frac{\sinh[k_0(\ell - z)]}{\sinh(k_0 \ell)}. \tag{4.2}$$

The planar-front approximation is an intermediate computational closure: it should not be mistaken for an exact solution satisfying $U(z_F(r,t),r,t) = 0$ at a curved front. Terms from the front displacement enter upon Taylor-expanding $U(z_F,r,t)$ about $z = \ell$.

5 Illustrative lowest-mode front evolution

For a steep front, the approximate axial energy balance is

$$\rho e \, \partial_t z_F = -\frac{4\sigma}{3\rho\kappa} \, \partial_z U|_{\text{front}} \,, \tag{5.1}$$

where e is the specific energy required to heat initially cold foam. Projecting the lowest-mode gradient of (4.2) onto the lowest mode, and taking the *model* front $z_F^{\text{mod}}(r,t) = c_0(t) J_0(k_0 r)$ with the auxiliary plane $\ell = c_0$, yields

$$\dot{c}_0 = \frac{D_M(t) b_0 k_0}{2 \sinh(k_0 c_0)}, \qquad D_M(t) = \frac{8\sigma U_s(t)}{3 e \rho^2 \kappa}. \tag{5.2}$$

This is a leading-mode approximation; it is not an exact pointwise moving-front solution. With $c_0(0) = 0$, its integral is

$$\boxed{c_0(t) = \frac{1}{k_0} \operatorname{arcosh}\left[1 + \frac{b_0 k_0^2}{2} \int_0^t D_M(s)\, ds\right].} \tag{5.3}$$

For a constant drive, D_M is constant and the integral equals $D_M t$. At leading order $b_0 \simeq 1$; retaining b_0 gives the source-normalization correction, $b_0 = 1 + \varepsilon/4 + O(\varepsilon^2)$, not an independently established $1 + \varepsilon/2$ correction. Higher-order consistency requires retaining higher radial modes and curved-front terms.

6 Calibration to the correct on-axis longitudinal position

The front may be predicted inaccurately along z even if the transverse *bending* is useful. To avoid confusing the correct on-axis position with the modal estimate, define

$$Z_F(t) = \text{correct or externally supplied on-axis front position}, \qquad z_F^{\text{mod}}(r,t) = \sum_{n=0}^{\infty} c_n(t) J_0(k_n r). \tag{6.1}$$

As $J_0(0) = 1$, $z_F^{\rm mod}(0,t) = \sum_n c_n(t)$. A *pure longitudinal correction*, preserving the original transverse displacement relative to the axis, is

$$\boxed{\widetilde{z}_F(r,t) = Z_F(t) + \sum_{n=0}^{\infty} c_n(t)\,[J_0(k_n r) - 1]\,.} \tag{6.2}$$

Indeed $\widetilde{z}_F(0,t) = Z_F(t)$. This is equivalent to translating the model front by $Z_F - \sum_n c_n$. It is a geometric calibration, not a new solution of (5.1) or of the complete nonlinear boundary-value problem. It also does not, by itself, correct the underlying temperature field.

6.1 On-axis curvature and curvature radius

For the meridional curve $z = \widetilde{z}_F(r,t)$, the unsigned curvature is

$$\mathcal{K}_m(r,t) = \frac{|\partial_r^2 \widetilde{z}_F|}{[1 + (\partial_r \widetilde{z}_F)^2]^{3/2}}. \tag{6.3}$$

By axisymmetry, $\partial_r \widetilde{z}_F(0,t) = 0$. Using

$$J_0(k_n r) = 1 - \frac{k_n^2 r^2}{4} + O(r^4), \tag{6.4}$$

one obtains (assuming sufficient convergence to differentiate the modal series)

$$\widetilde{z}_F(r,t) = Z_F(t) - \frac{r^2}{4R^2} \sum_{n\geq 0} c_n(t)\lambda_n^2 + O(r^4), \tag{6.5}$$

$$\partial_r^2 \widetilde{z}_F\Big|_{r=0} = -\frac{1}{2R^2} \sum_{n\geq 0} c_n(t)\lambda_n^2. \tag{6.6}$$

Therefore the *on-axis meridional curvature radius* is

$$\boxed{\mathcal{R}_c(t) = \frac{2R^2}{|\sum_{n=0}^{\infty} c_n(t)\lambda_n^2|}.} \tag{6.7}$$

At the axis, the two principal curvatures of the smooth surface of revolution agree in magnitude; away from the axis the meridional and azimuthal principal curvatures generally differ. Most importantly, (6.7) is *independent* of $Z_F(t)$ for an additive longitudinal correction.

6.2 Fundamental mode and alternative rescaling

Keeping only $n = 0$ in (6.2),

$$\widetilde{z}_F(r,t) \simeq Z_F(t) + c_0(t)[J_0(k_0 r) - 1], \qquad \boxed{\mathcal{R}_c(t) \simeq \frac{2R^2}{|c_0(t)|\lambda_0^2} \simeq \frac{R^2}{\varepsilon |c_0(t)|}.} \tag{6.8}$$

If instead one decides to rescale the *entire fundamental-mode amplitude* to match Z_F, the different assumption is

$$\widetilde{z}_F^{\rm scale}(r,t) = Z_F(t) J_0(k_0 r), \qquad \boxed{\mathcal{R}_c^{\rm scale}(t) = \frac{2R^2}{|Z_F(t)|\lambda_0^2} \simeq \frac{R^2}{\varepsilon |Z_F(t)|}.} \tag{6.9}$$

The two formulas are *not interchangeable*: translation preserves curvature; amplitude rescaling changes it. A more general amplitude adjustment $A(t)$ multiplying the bracketed sum in (6.2) scales the on-axis curvature by $|A(t)|$.

7 Physical interpretation and limitations

The cylindrical eigenfunctions and Robin eigenvalues determine the radial dependence associated with wall losses. The leading eigenvalue $k_0 R \sim \sqrt{2\varepsilon}$ produces the first-order reduction of axial penetration toward the sleeve. The auxiliary solution (4.1) and the illustrative evolution (5.3) approximate the nonlinear moving-front problem; higher-mode coupling and the actual curved-front boundary condition should be solved to obtain a controlled higher-order correction. The calibration (6.2) is appropriate when a separate model or measurement supplies a more reliable *on-axis* trajectory and one wants to retain the Bessel-predicted bending.

8 Selected Bessel identities

$$J_0'(x) = -J_1(x), \qquad \frac{d}{dx}[x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x), \tag{8.1}$$

$$J_\nu'(x) = \tfrac{1}{2}[J_{\nu-1}(x) - J_{\nu+1}(x)], \qquad J_0(x) = 1 - \frac{x^2}{4} + O(x^4). \tag{8.2}$$

The norm (3.8) is the appropriate Robin-boundary orthogonality formula for this problem.

9 Wall albedo: definition versus inference

The wall albedo is the fraction of incident radiative flux re-emitted or reflected back into the foam, in the effective boundary model:

$$a = \frac{F_{\text{returned}}}{F_{\text{incident}}}, \qquad 0 \le a \le 1, \qquad F_{\text{net loss}} = (1-a)F_{\text{incident}}. \tag{9.1}$$

Consequently the characteristic Robin parameter determines only the combination

$$1 - a = \frac{4\varepsilon}{3\rho\kappa R}. \tag{9.2}$$

Measured outgoing temperatures and an independently estimated wall heating rate are not, by themselves, sufficient to justify an algebraic ratio such as $(F_0 + \dot{E}_w/2)/(F_0 - \dot{E}_w/2)$ for albedo. A detector/transport energy balance defining which flux is incident and returned is required before inferring $a(t)$. If a is time dependent, the eigenbasis is also time dependent and the constant-ε modal derivation needs revision.

References

[1] O. A. Hurricane and J. H. Hammer, "Bent Marshak waves," *Physics of Plasmas* **13**, 113303 (2006), doi:10.1063/1.2388268.

[2] J. H. Hammer and M. D. Rosen, *Physics of Plasmas* **10**, 1829 (2003).